

₹ 500,000 to ₹ 2 million. Solutions like these can be quite cost-effective and beneficial for police administratives in India.

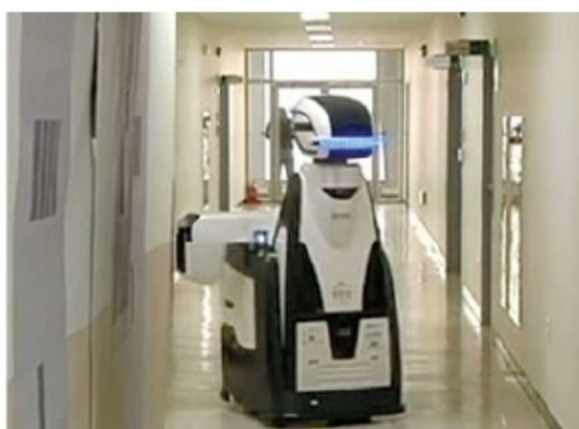
On the other hand, police departments in other countries, especially in Asia, have already made progress in utilising robots. For instance, since 2011, South Korea has been using robotic guards to invigilate their prisons. Started as a trial project at a prison in Pohang, adaptation of these automated prison guards has expanded across the region.

The self-patrolling robots can sense inmate behaviour patterns and surrounding conditions to trigger alarms and notify prison administration in case of problems. These also facilitate two-way communication.

However, the robots are not built to engage physically by themselves. Their implementation has substantially reduced the load on manual prison patrols and has slashed down in-prison incidents over the years.

Taiwan police recently deployed robots for citizen assistance purposes. Ayuda robot, built by Syscom Computer Engineering, can file and print formal police complaints for citizens, run citizen ID verification, provide road assistance and more. It can also be used for surveillance and security purposes powered by the various technologies lying underneath the design.

Max Yang, executive director, Syscom Computer Engineering, says, "The robot combines multiple sen-



Robots patrolling a prison in South Korea
(Credit: www.phys.org)

sors and utilises techniques like facial recognition for citizen identification, natural language processing-based speech for interaction and computer vision for surrounding situation monitoring. Inbuilt microphone facilitates two-way communication."

Ayuda's body is made of acrylonitrile-butadiene-styrene (ABS). It is 1.6-metre tall and weighs 100 kilograms. Using a lithium battery, it can run for eight hours, following four hours of charge. Taiwan police will deploy more such robots in 2019.

Dubai Police Department grabbed eyeballs in 2017 when it launched robotic officers to reduce manual load in crowded areas. The 1.67-metre tall robots are tasked with real-time surveillance and citizen support near malls, stations, hospitals and the like. People can use the touchscreen on the machines to access road directions, pay fines, lodge complaints and more.

Dubai police is also using its facial recognition capabilities to identify individuals. A live video feed keeps the control centre updated regarding the areas that the robot covers. It is used to detect parking violations too and can pursue subjects up to a speed of 80kmph. The police department plans to have 25 per cent of their workforce as robots by 2030, without replacing or reducing human requirements in the force.

Proper maintenance of robots can ensure great advantages for India's decision-makers in strengthening law and order in the country. As many modern technology firms are delving deep to create such solutions, it is a good time for the Indian police force to explore these options and leverage on these technologies.



Ayuda robot

Artificial Intelligence: The Harbinger Of New-Age Banking

➤ **AI platforms are helping banks identify loopholes in their networks, prevent cyber-attacks, improve customer support and optimise overall operations**

Indian banks have been on their toes since digital systems started to evolve and become ever-pervasive. To help them improve and expand, banks have turned to a modern technology, artificial intelligence (AI). AI, along with other correlated techniques like data analytics and machine learning are helping them improve customer experience and support, prevent breaches, detect vulnerabilities, reduce fraudulent transactions and more. Eventually, this is leading to lower operational expenses, better business and a renewed essence of banking for customers. In this article, we look at how AI and similar technologies are making this possible.

AI for customer support

Some of India's largest banks have adopted AI to serve their customers faster and more accurately. For instance, State Bank of India (SBI) deployed its chatbot SIA (which is short for SBI Intelligence Assistant), which takes care of inquiries over its website. It provides services such as answering queries, suggesting loans, helping in opening an account, making deposits, registering users and much more.

The chatbot can respond to 10,000 queries per second, which takes sub-